GuitarTap

Tap Tone Analysis Report

March 9, 2026
3:06 PM

Location:	Plate Classical
Type:	Material (Plate)
Notes:	Umik-1
Frequency Range:	30.0 Hz – 600.0 Hz
Microphone:	Umik-1 Gain: 18dB · uncalibrated

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
54.8 Hz	-67.7 dB	A1	2.6	FLC (Diagonal)
67.4 Hz	-56.1 dB	C#2	11.5	Longitudinal (L)
117.5 Hz	-42.4 dB	A#2	20.0	Cross-grain (C)

Three-Tap Measurement Process:

- **1. Longitudinal (L) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (C) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. FLC (Diagonal) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.

Material Properties

Sample Dimensions

Length: **557.5 mm** Width: **220.5 mm** Thickness: **4.85 mm**
Mass: **208.0 g** Density: **0.349 g/cm³**

fL: **67.4 Hz** fC: **117.5 Hz** fLC: **54.8 Hz**

Speed of Sound (L): **4205 m/s**
Speed of Sound (C): **1147 m/s**
Young's Modulus (L): **6.17 GPa**
Young's Modulus (C): **0.46 GPa**
GLC (Shear Modulus): **0.818 GPa**

Specific Modulus (L): **17.7** (Fair)
Specific Modulus (C): **1.3** (Very Good)
Radiation Ratio (L): **12.1**
Radiation Ratio (C): **3.3**

Cross/Long Ratio: **0.074**
typical: 0.04 – 0.08

Long/Cross Ratio: **13.4**
typical: 12 – 25

Overall Quality: Good

Gore Target Thickness

2.72 mm

Body: 490 × 390 mm · f_vs = 60 (Classical Top)
GLC (Shear Modulus): 0.818 GPa